

Multiclass Skin Disease Classification within Dermoscopic Images Using Deep Neural Networks

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ABSTRACT

Computer-aided skin lesion classification has been gaining attention in dermoscopy and skin cancer diagnosis, as early detection reduces the complexity of the treatment process. Various techniques that utilize the power of deep neural networks have been found in recent years to tackle this problem. In this study, we evaluate the effects of different preprocessing methods as well as classifiers on the multiclass categorization of seven skin diseases, with InceptionV3 being used as our backbone. These methods are evaluated using the ISIC 2018 Task 3 dataset, on which we achieve 76.32% accuracy after employing segmentation with dilating and cropping on our training dataset. This study reveals that image preprocessing properly can enhance the models' performance in classifying skin diseases, and deep learning has great potential for assisting human experts in repetitive tasks.

CCS CONCEPTS

• Computing methodologies → Image segmentation; Neural networks; • Applied computing → Imaging.

KEYWORDS

ISIC, dermoscopy, computer-aided, classification, skin disease, preprocessing

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1 INTRODUCTION

Skin cancer poses a significant public health challenge, with more than five million newly diagnosed cases occurring annually in the United States [23]. The increase in skin cancer is a consequence of the reduction in the ozone layer, which protects us against ultraviolet rays [20]. Among numerous types of skin cancer, melanoma is responsible for an overwhelming majority of skin cancer deaths [33]. In 2020, roughly 6850 new death cases due to melanoma were recorded in the US [26]. This indicates that melanoma ranks among the most lethal types of skin cancer, and there is a possibility of a rise in the mortality rate caused by melanoma infections over the upcoming decades.[21].

Early detection and accurate diagnosis of melanoma are essential to the patient, as the survival rate can reach 90% in early detected lesions [29]. Several methods can be applied to early detection of melanoma as well as other types of skin cancer. As pigmented lesions are visible on the skin's surface, expert visual inspection can detect melanoma at an early stage. However, manual inspection by dermatologists is often a tedious, time-consuming, and objective job. Such a visual assessment via the naked eye may create a high risk of errors, resulting from the comparability between the lesions and healthy tissues [5]. With the help of image analysis, another method comes up to provide dermatologists with better diagnosis results. In dermatology, dermoscopic images, also known as epiluminescence light dermoscopy, is the most common modality used to analyze these pigmented skin lesions [19]. Dermoscopy produces images of skin lesions with more details, which is far better than examining the naked eye as it improves the accuracy of skin lesion classification up to 50% [34]. However, the diagnostic accuracy of dermatoscopy is still closely linked to the expertise of dermatologists [31].

To mitigate all the limitations mentioned earlier and improve skin cancer diagnosis accuracy, computer-aided diagnosis (CAD) has been introduced. CAD leverages the power of machine learning

algorithms and image analysis to assist dermatologists in classifying skin cancer based on images [22]. By automating the analysis process, CAD systems aim to improve diagnostic accuracy, reduce inter-observer variability, and expedite the identification of malignant lesions. Additionally, significant progress has been made in developing CAD systems for skin cancer classification in recent years. These systems utilize large datasets of dermoscopic images, incorporating features such as asymmetry, border irregularity, color variation, and structural patterns to differentiate between benign and malignant lesions [7]. Machine learning algorithms, including support vector machines, artificial neural networks, and deep learning models, have demonstrated remarkable performance in automating the classification process [10, 24, 32, 35]. However, challenges remain in achieving high sensitivity, specificity, and generalizability across diverse populations and skin types.

In this paper, we evaluate an approach that leverages a convolutional neural network (CNN) architecture as the backbone, with various preprocessing methods on our dataset. Furthermore, because of its effectiveness in classification tasks, we assess the power of Support Vector Classifiers (SVCs) when combined with a CNN feature extractor. This approach achieves the highest accuracy of 76.32% accuracy on the skin lesion classification task. Taking into consideration the factors above, we outline our primary contributions in this paper as follows:

- Utilize the pre-trained weights from the Inception-V3 model as initialization and retrain it with our dataset on the entire neural network. The network here is used as an automatic feature extractor.
- Train 2 hybrid CNN-SVCs along with traditional CNN-Softmax model.
- Use the weighted focal loss method to solve the problem of heavy class imbalance.
- Evaluate several data-processing techniques on the training dataset, including augmentation, hair removal, and dilated cropped segmentation.

The following is the structure of this paper: First, we analyze the existing literature and explore the advantages and disadvantages of current approaches in Section II. After discovering related works, in Section III, we show a brief explanation of our proposed methods, as well as describe the datasets that we used. In Section IV, we demonstrate how we conduct experiments and show results, and last but not least, we will discuss potential developing directions in Section V. We hope that our research contributes to the ongoing efforts to enhance early detection and accurate diagnosis of skin cancer, ultimately leading to improved patient outcomes and reduced mortality rates.

2 RELATED WORK

2.1 Preprocessing

Hasan et al. [11] suggested geometry- and intensity-based image augmentations such as gamma, logarithmic, and sigmoid corrections. Besides, class rebalancing methods, including the combination of other dermoscopy datasets and a weighted loss function, have helped to tackle overfitting and bias problems.

Mahbod et al. [18] explored the effects of skin lesion segmentation masks on dermoscopic image classification, using either

manually or automatically created segmentation masks. This work showed that a dilated cropping strategy significantly improved classification performance, which suggested this is the best way to use a segmentation mask.

Hoang et al. [13] proposed a new segmentation approach using entropy-based weighting and first-order cumulative moment (EW-FCM), which provided better segmentation accuracy than the original Otsu [9] and entropy weighting (EW) [30] scheme.

Singh et al. [27] introduced a novel segmentation model called Bayesian MultiResUNet, designed to estimate uncertainty in predicted segmentation maps. This model represents an enhanced iteration of U-Net, incorporating MultiRes blocks and Res path structures as sources of inspiration. Experimental results have demonstrated the superior performance of the proposed Bayesian MultiResUNet over alternative segmentation models.

Khan et al. [16] proposed a segmentation method using the most discriminant deep features. To begin, the input images undergo an initial enhancement process using local color-controlled histogram intensity values (LCcHIV). Subsequently, we estimate saliency through a newly developed Deep Saliency Segmentation technique, which employs a specialized convolutional neural network (CNN) of ten layers.

Bencevic et al. [6] propose an innovative approach named Polar Res-U-Net++. This method utilizes a polar transformation to convert Cartesian space images into the polar domain, leading to reduced image dimensionality and separation of segmentation and localization tasks. Consequently, the network's convergence is improved. The effectiveness of this approach is evident in its successful segmentation of liver, polyp, skin lesions, and epicardial adipose tissue.

2.2 Feature extraction

He et al. [12] propose a Fully Transformer Network (FTN) to learn long-range contextual information for skin lesion analysis. FTN is a hierarchical Transformer computing feature using Spatial Pyramid Transformer (SPT). SPT has linear computational complexity as it introduces a spatial pyramid pooling (SPP) module into multi-head attention (MHA) to largely reduce computation and memory usage. After conducting experiments, it is shown that FTN consistently outperforms other state-of-the-art CNNs regarding computational efficiency and the number of tunable parameters.

Adegun et al. [1] proposed a framework that consists of two stages: The first stage uses an encoder-decoder Fully Convolutional Network (FCN) with skip pathways, combining long and short-cut connections to improve residual learning and training effectiveness. The second stage proposes a novel FCN-based DenseNet framework and employs hyperparameters optimization techniques for better network complexity computing efficiency. With those techniques, they achieved 98% accuracy, 98.5% recall, and 99% of AUC score, respectively on HAM10000.

Ali et al. [4] propose a deep convolutional neural network (DCNN) model based on a deep learning approach for accurately classifying benign and malignant skin lesions. This model was evaluated using the HAM10000 dataset, resulting in the highest training accuracy of 93.16% and testing accuracy of 91.93%. The model is indicated

to be more reliable and robust when compared with existing transfer learning models such as AlexNet, ResNet, VGG-16, DenseNet, MobileNet, etc.

Srinivasu et al. [28] proposed a computerized process of classifying skin disease through deep learning-based MobileNet V2 and Long Short Term Memory (LSTM), collaborated with a grey-level co-occurrence matrix which is used for assessing the progress of diseased growth. The work eventually resulted in more than 85% accuracy on the HAM10000 dataset.

Hosny et al. [14] proposed a method that utilized transfer learning with pre-trained AlexNet. This led to exceptional performance, with impressive percentages of 98.70% for accuracy, 95.60% for sensitivity, 99.27% for specificity, and 95.06% for precision. These results highlight the remarkable capabilities of AlexNet in achieving highly accurate and precise predictions.

Hasan et al. [11] proposed a classification framework called Dermo-Expert, which was a combination of pre-processing and hybrid-CNN. The hybrid-CNN fused three different feature extractor modules to produce output feature maps, which were fed to distinct fully connected layers for ensembling.

Hoang et al. [13] introduced a two-dimensional wide-ShuffleNet network, which is composed of the ShuffleNet units and the idea of skip connections. The proposed lightweight method was proven to reduce the number of parameters while preserving higher accuracy.

Singh et al. [27] developed a novel deep architecture called SkinNet, which was able to address the black-box property of traditional models and raise the prediction's confidence. Monte Carlo dropout and test time augmentation techniques were utilized in this method to approximate epistemic and aleatoric uncertainty. Moreover, using saliency-based methods such as XRAI, Grad-CAM, and Guided Backprop, post-hoc explanations for predictions were provided.

Mahbod et al. [17] explored the effect of using six different image sizes ranging from 224×224 to 450×450 and three well-established CNNs including EfficientNetB0, EfficientNetB1 and SeReNeXt-50. A multi-scale multi-CNN (MSM-CNN) fusion approach based on a three-level ensemble strategy was also suggested, using three network architectures trained on images of multiple scales. This research proved that image cropping performed better than image resizing at all evaluated image scales. On the ISIC2018 test set, the MSM-CNN yielded a balanced multi-class accuracy of 86.2%.

2.3 Feature selection and Classifier

Khan et al. [16] used a multiset maximum correlation analysis (MMCA) and Kernel Extreme Learning Machine (KELM) as a classifier.

Afza et al. [2] introduced a computationally efficient method, which included the selection of best features using hybrid whale optimization and entropy-mutual information (EMI) approach, a fusion of selected features using the modified canonical correlation-based approach, and finally, an extreme learning machine-based classifier. The achieved accuracy on the ISIC2018 dataset was 94.36%.

Agarap et al. [3] emulated an architect using a simple 2-Convolutional Layer with a Max Pooling model, combined with a linear SVM for image classification. They achieved 90.72% while the sophisticated CNN-Softmax only reached 91.86%.

3 PROPOSED METHODS

3.1 Overview

The methods we evaluate, as shown in Figure 9, consist of two main components: (1) preprocessing dataset and (2) evaluating the efficiency of two types of classifiers on the Inception-V3.

In the initial stage of our research, our primary objective is to optimize the quality of the input data. This goal is achieved by applying several advanced techniques to our training dataset. Firstly, a hair removal process is implemented to effectively mitigate noise and significantly enhance the clarity of dermoscopic images. Secondly, we employ image augmentation, a methodology that encompasses a diverse range of transformations applied to the images. This augmentation process enriches the dataset, introducing a wider spectrum of training examples, thereby potentially augmenting the model's overall performance. Additionally, we leverage the potential of image segmentation to help the model concentrate exclusively on the most critical and relevant features of the lesions during the learning process. Such a comprehensive approach to data enhancement lays a strong foundation for the subsequent stages of our scientific investigation.

Moving on to the second stage, after training the pre-trained Inception-V3 with the preprocessed datasets, we evaluate its performance using three classifiers: a normal softmax classifier, a linear support vector classifier, and a support vector classifier with RBF kernel. The softmax classifier applies a probability distribution over multiple classes, allowing us to classify the input data into different categories. On the other hand, the support vector classifier combined with Inception-V3 architecture will together create a hybrid CNN-SVM model. It uses hyperplanes to separate the data points into distinct classes.

Then, by comparing the results obtained from both classifiers, we can determine which method yields superior performance. We select the approach that achieves higher accuracy or provides better overall results and use it to train the final model. This comparative analysis ensures that we choose the most effective classifier for our specific task, thereby optimizing the performance of our proposed method.

3.2 Preprocessing

3.2.1 Hair removal. Effective noise control plays a crucial role in image-based problems since noise can significantly disrupt the learning process of neural networks. In the case of skin images, one of the most problematic types of noise is human hair. To address this issue, an open-source software called Dull Razor is utilized to remove hairs from images automatically. Initially, the software identifies the hair's location within the image. Subsequently, it smoothens the affected areas and replaces the hair by employing adaptive median filtering on neighboring pixels[15].

Nevertheless, the robustness of the neural network can be improved by distributing suitable amounts of noise in the images [25]. For this reason, we apply the Dull Razor algorithm to remove all hair in the dataset.

3.2.2 Augmentation. Image augmentation is a useful method to improve data diversity when the raw dataset alone is not enough[25].

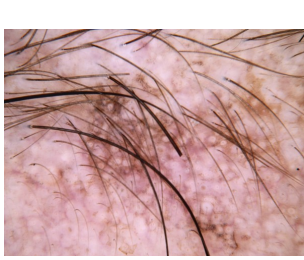


Figure 1: Original image

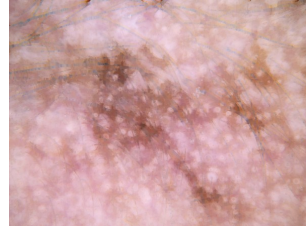


Figure 2: Hair removed image

In our work, we do a range of transformations to the images, including brightness adjustment, rotation, flipping, shearing, and zooming, with a level of randomness.

3.2.3 Segmentation. We apply the proposed method of Bencevic et al. [6] based on the polar transformation of images. In their method, two neural networks were used, including a center-point predictor that predicts a heatmap from an input image, which is then used to produce polar origin for polar transformation, and a trained polar network for producing a binary mask output. Eventually, the segmentation label is transformed back to Cartesian coordinates. After the mask is produced, the lesion inside the mask is dilated by a factor of 1.4 and then cropped in order to retain images' background information.

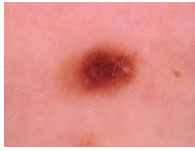


Figure 3: Original image

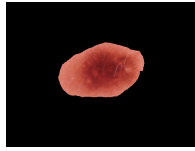


Figure 4: Segmentation mask



Figure 5: Dilated cropped image

3.3 Technique

We design three models with different classifiers, but both use the Inception-v3 architecture as a CNN backbone. Inception-v3 is a famous 48-layer convolutional neural network that belongs to the Inception family and was developed by Google for image classification and object recognition tasks. The key idea behind the Inception-v3 model is to use multiple parallel convolutional layers of different sizes within the same module, called an "Inception module." These modules are stacked on top of each other, forming a deep network. To be more detailed, basic features are extracted by a few convolutional layers and max pooling operations and then are passed through multiple Inception modules. This allows the model to capture information at various scales and levels of abstraction simultaneously, leading to better representation learning.

In addition to the parallel convolutional layers, Inception-v3 also incorporates other techniques to improve performance and efficiency. These include the use of 1x1 convolutions, which reduce the dimensionality of the input and enable more efficient computations, and the addition of auxiliary classifiers, which are small classifiers

inserted at intermediate layers to provide additional gradients during training and combat the vanishing gradient problem. 6 sketches the architecture of the Inception-v3 model.

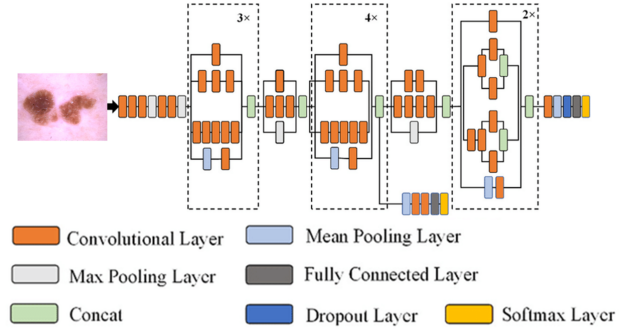


Figure 6: The architecture of Inception-v3 model.

Source: <https://arxiv.org/abs/1512.00567v3>

First, the pre-trained Inception-v3 is utilized with weights initialized from the ImageNet [8] dataset. It is combined with additional layers, including a softmax output layer to produce classification results. The whole network is trained using our training dataset, which is previously mentioned as a combination of ISIC 2018 and 2019 datasets. After training the original layers of Inception-v3 as a feature extractor, we feed the feature vectors to different classifiers:

- (1) **Softmax classifier.** We add three more fully connected layers, including a 3-class softmax layer at the end of the network. Feature vectors extracted are linearly combined to produce three outputs corresponding to 3 classes. These 3 numbers z_0, z_1, z_2 are then passed into this formula:

$$y_i = \frac{e^{z_i}}{\sum_{j=1}^3 e^{z_j}} \quad (1)$$

where y_i can be interpreted as the probability that an image is predicted to be of class i .

- (2) **Linear support vector classifier.** This is a machine learning algorithm used for regression and classification problems. Regarding classification problems, its mechanics is to find the most ideal hyperplane that can linearly separate classes by choosing points (a.k.a support vectors) closest to the hyperplane, which will maximize the margin.
- (3) **Radial basis function support vector classifier.** When boundaries between classes are non-linear, SVC applies the kernel trick. It maps our input data points from low-dimensional space to high-dimensional space, where data points will be linearly separated. In our models, we use radial basis function (RBF) as a kernel, the most preferred one in most non-linear cases. Figure 7 describes how different kernels separate classes:

To select the best hyperparameters are selected using the Grid Search algorithm to provide the best result.

To summarize, our general method is visualized in figure 8.

For training and evaluation, we use focal loss, a dynamically scaled cross-entropy loss that can gradually decrease to zero when

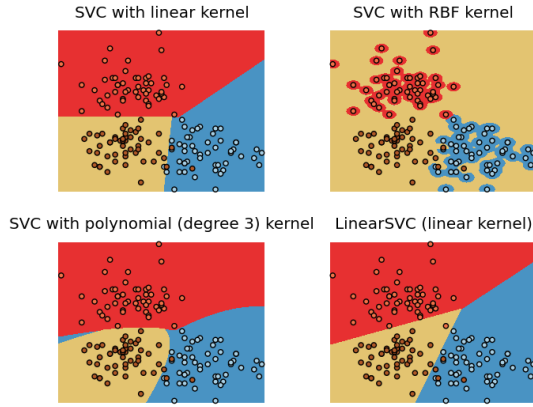


Figure 7: Visualization of classification using SVCs with different kernels.

Source: <https://stackoverflow.com/questions/17794313/unexpected-results-when-using-scikit-learns-svm-classification-algorithm-rbf-k>

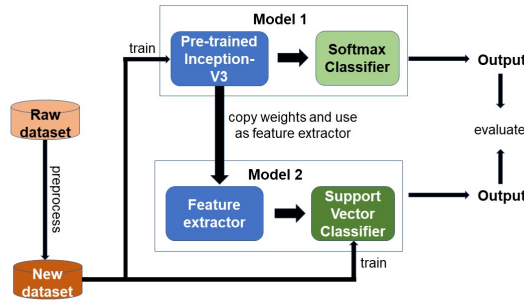


Figure 8: The architecture of our proposed model.

confidence in the correct class increases. To tackle the problem of heavy class imbalance, we also add weights to our loss function. Mathematically, these weights are automatically adjusted inversely proportional to class frequencies in the input data. Therefore the weight of dominant classes during training is scaled down, whereas rare classes are more focused. Below is the formula for our proposed loss function:

$$WFL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

where α_t is the weight of class, p_t is the probability that the training example will be correctly predicted, and the focusing hyperparameter $\gamma > 0$ allows hard-to-classify examples to be penalized more heavily relative to easy-to-classify examples.

4 EXPERIMENTAL RESULTS

4.1 Implementation details

We conduct our experiments by implementing parts of the following Python libraries: **TensorFlow**, **scikit-learn**, and **Keras** on Jupyter notebooks. Our models utilize the InceptionV3 architecture as the backbone, pre-trained on the ImageNet dataset and accessed through Keras APIs. It transforms image inputs of shape 299x299x3 into feature vectors. Additionally, to create a hybrid CNN-SVM model, we incorporate the C-Support Vector classifiers provided by scikit-learn.

In our initial experiments, we set the learning rate to 3e-4. However, due to fluctuations observed in the training and validation loss, we decided to reduce the learning rate to 1e-4. The batch size for each training step is set to 128. To ensure sufficient training, our models undergo at least 40 epochs. After the 40th epoch, we employ early stopping, halting the training process when the validation loss ceases to improve.

We conduct our experiments on **Kaggle Kernels**, utilizing the GPU P100 as an accelerator. This setup provides us with 15.9GB of GPU memory and 13GB of RAM, enabling efficient execution of the training sessions.

4.2 Dataset

ISIC Challenges are renowned annual competitions focusing on computer-aided diagnosis and melanoma detection using dermoscopic images. The dataset used in this report is sourced from the ISIC 2018 Task 3 dataset. Our training and validation set contains 8010 and 2005 images, respectively, achieved after performing an 80-20 train-test split on the ISIC original training set. On the other hand, the test set with 1512 is used for testing. These datasets belong to 7 classes and consist of a vast collection of high-resolution images of various skin lesions acquired from diverse clinical settings and imaging devices. The dataset mentioned above has an issue with the unequal distribution of images among different classes. To solve this issue, a weighted focal loss was used to ensure that each class has an equal impact on the result. This improves the reliability and generalizability of the findings.

Table I, Table II and Table III show the number of images belonging to each of 7 classes, respectively in 3 datasets: respectively in 3 sets: training, validation and test set.

Table 1: Training set

Class labels	No. of images	Percentage
AKIEC	261	3.26%
BCC	411	5.13%
BKL	879	10.97%
DF	92	1.15%
MEL	890	11.11%
NV	5364	66.97%
VASC	113	1.41%

Table 2: Validation set

Class labels	No. of images	Percentage
AKIEC	66	3.29%
BCC	103	5.14%
BKL	220	10.97%
DF	23	1.15%
MEL	223	11.12%
NV	1341	66.88%
VASC	29	1.45%

Table 3: Test set

Class labels	No. of images	Percentage
AKIEC	43	2.84%
BCC	93	6.15%
BKL	217	14.35%
DF	44	2.91%
MEL	171	11.31%
NV	909	60.12%
VASC	35	2.31%

4.3 Evaluation metrics

Confusion Matrix: The Confusion Matrix shows the number of True Positives (TP), False Positives (FP), True Negatives (TN) and False Negatives (FN) in a compact manner.

Classification Report: The different models are evaluated by calling the ‘classification_report’ function from scikit-learn. This function provides evaluation with three metrics: accuracy, macro average, and weighted average.

4.3.1 Accuracy.

Accuracy is a straightforward metric that measures the proportion of correctly classified samples from the total number of samples in the dataset. The accuracy metric is defined as:

$$Accuracy = \frac{TP}{TP + TN + FP + FN} \quad (3)$$

4.3.2 Macro-Averaged F1 Score.

The F1 score is the harmonic mean of precision and recall and is a useful metric for the comprehensive assessment of any classification approach. In the macro-averaged F1 score, we calculate the F1 score for each class individually and then take the average. This metric evaluates the model’s ability to classify all classes, regardless of their sizes, correctly.

4.3.3 Macro-Averaged Recall, or Balanced Multiclass Accuracy.

The macro-averaged recall, also known as balanced multiclass accuracy, calculates the recall for each class separately and then takes the mean of these recall scores. This metric is particularly useful when you want to ensure that the model performs well for each class without being biased by the class distribution. This is also the main metric for evaluation at Task 3 of the ISIC competition 2018.

4.3.4 Weighted-Averaged F1 Score.

In the weighted-averaged F1 score, we calculate the F1 score for

each class, just like in the macro-averaged F1, but then we take a weighted average where the class distribution determines the weights. This metric addresses the imbalance of our dataset, as it considers the importance of each class based on its prevalence in the dataset.

4.4 Experiment setups and results

In the first phase, we compare two different approaches to training: training from scratch vs transfer learning. We only use 3 (AKIEC, BCC, BKL) per 7 classes of our datasets for faster primary evaluation in this phase. To train the model from scratch, pre-trained weights of Inception-V3 are used as initialized weights. On the other hand, transfer learning involves freezing the feature extractor for and only training the classifier at the top of our model. For each approach, two classifiers, softmax and linear-SVC, are employed to provide a more comprehensive assessment of the two mentioned approaches. Evaluation of our test dataset is shown in Table IV. It can be observed that training from scratch gives significantly higher results than transfer learning.

Table 4: Comparison between training from scratch and transfer learning

Approach	Classifier	Accuracy	Macro F1	BMA	Weighted F1
Train from scratch	Softmax	0.76	0.64	0.62	0.74
Train from scratch	Linear SVC	0.76	0.65	0.63	0.75
Transfer learning	Softmax	0.62	0.44	0.45	0.59
Transfer learning	Linear SVC	0.60	0.50	0.51	0.60

In the second phase, we trained the models exclusively from scratch. Our baseline comprises the Inception-V3 architecture with a standard softmax classifier and the original dataset, which has not been through any preprocessing. Our primary objective was to compare different setups of benchmark models comprising three types of preprocessing techniques and three classifiers: softmax classifier, linear SVC, and RBF SVC. The evaluation results of different model setups, validated using four metrics (accuracy, macro F1-score, BMA, and weighted F1-score), are presented below.

Table 5: Evaluation of different models, metric: accuracy.

Dataset/Classifier	Softmax	Linear SVC	RBF SVC
Original	0.7460	0.7474	0.7282
Original + Augmented	0.7606	0.7599	0.7593
Original + Hair-removed	0.7480	0.7526	0.7474
Segmented (dilated & cropped)	0.7606	0.7632	0.7546

Table 6: Evaluation of different models, metric: balanced multiclass accuracy (BMA)

Dataset/Classifier	Softmax	Linear SVC	RBF SVC
Original	0.42	0.58	0.61
Original + Augmented	0.59	0.58	0.60
Original + Hair-removed	0.49	0.56	0.59
Segmented (dilated & cropped)	0.55	0.61	0.58

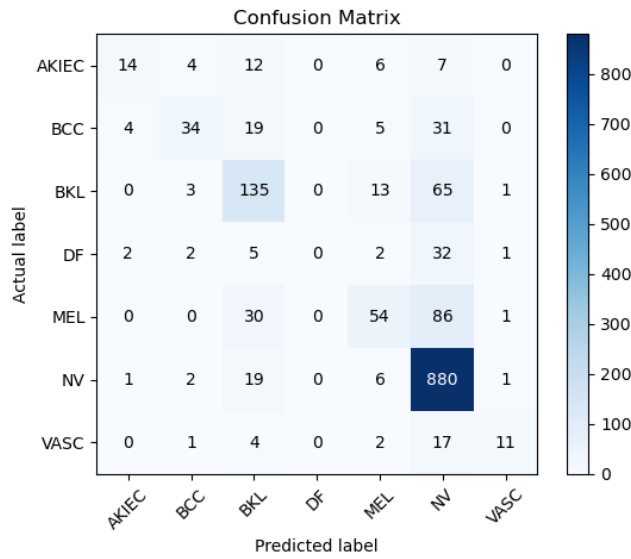


Figure 9: Confusion matrix for our baseline model.

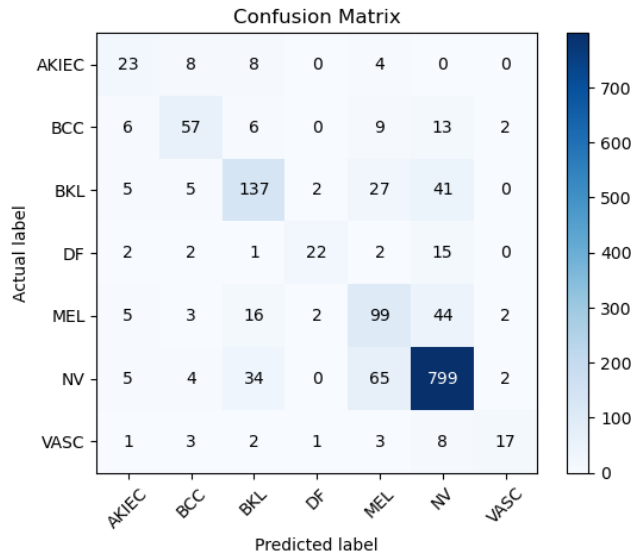


Figure 10: Confusion matrix for our best performance model.

Table 7: Evaluation of different models, metric: macro F1-score

Dataset/Classifier	Softmax	Linear SVC	RBF SVC
Original	0.55	0.61	0.60
Original + Augmented	0.60	0.60	0.61
Original + Hair-removed	0.55	0.59	0.60
Segmented (dilated & cropped)	0.60	0.63	0.62

Table 8: Evaluation of different models, metric: weighted F1-score

Dataset/Classifier	Softmax	Linear SVC	RBF SVC
Original	0.71	0.75	0.74
Original + Augmented	0.77	0.76	0.77
Original + Hair-removed	0.73	0.75	0.75
Segmented (dilated & cropped)	0.75	0.76	0.75

Table 9: Comparison of proposed methods at the ISIC 2018 Competition

Model	Method	Score (BMA)
Large Ensemble	Heavy multi-cropping and loss weighting	0.856
Top 10 Models Averaged	Segmented (dilated & cropped), augmented	0.885
Ensemble methods of SENet and PNASNet	Color Constancy, Data Augmentation	0.845
Densenet	Cropped lesion images	0.815
Inceptionv4, ResNet-152, and DenseNet-161	Class-weighted cross-entropy, Segmented (dilated & cropped)	0.803
ResNet-50/101/152	Augmented	0.786
Ensemble of Multiple Trained Classifiers	Augmented, hair-removed	0.785
VGG, InceptionV4, ResNet152, and NasNet	Random rotation and flipping	0.780

Regarding accuracy and balanced multiclass accuracy (BMA) from Table V and Table VI, the model trained with linear SVC on the segmented dataset exhibited the best performance, achieving an accuracy of 76.32% and BMA of 61%. A slight improvement was also observed when the augmentation and hair removal datasets were combined with the original samples. Regarding balanced accuracy, using models with SVCs clearly shows higher performance than only using a softmax layer. The results were consistent with the macro F1-score in Table VI, with the segmented dataset and linear SVC combination outperforming the baseline with a score of 0.63 versus 0.55. In terms of weighted F1-score, the augmentation method attained the highest score of 77%, surpassing all other techniques, as shown in Table VII. Additionally, upon comparing all three classifiers, the linear SVC consistently achieved the highest results in most cases. A comparison between our baseline model versus our best performance model is shown in figure 10 and 11, from where we can observe a rise in all metrics and especially less variance among seven classes' metrics.

5 CONCLUSION AND FUTURE WORK

In this paper, we have tested the effect of various methods of pre-processing and classifying on the performance of dermoscopic image classification and observed some interesting results. Our experiments suggest that these preprocessing methods, including augmentation, dilated segmentation, and hair removal, positively impact the model's categorization results. Appropriately using segmentation masks is suggested to enhance the model's prediction. Besides, it appears that the hybrid InceptionV3-SVC is a good alternative to InceptionV3-Softmax in most cases. Our approach, while not yet achieving results surpassing the latest updated methods, has demonstrated the potential of combining preprocessing methods

with a hybrid architecture on a more sophisticated backbone and shows great promise for further research in the field of dermoscopy.

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REFERENCES

- [1] Adekanmi A Adegun and Serestina Viriri. 2020. FCN-based DenseNet framework for automated detection and classification of skin lesions in dermoscopy images. *IEEE Access* 8 (2020), 150377–150396.
- [2] Farhat Afza, Muhammad Sharif, Muhammad Attique Khan, Usman Tariq, Hwan-Seung Yong, and Jaehyuk Cha. 2022. Multiclass skin lesion classification using hybrid deep features selection and extreme learning machine. *Sensors* 22, 3 (2022), 799.
- [3] Abien Fred Agarap. 2017. An architecture combining convolutional neural network (CNN) and support vector machine (SVM) for image classification. *arXiv preprint arXiv:1712.03541* (2017).
- [4] Md Shahin Ali, Md Sipon Miah, Jahurul Haque, Md Mahbubur Rahman, and Md Khairul Islam. 2021. An enhanced technique of skin cancer classification using deep convolutional neural network with transfer learning models. *Machine Learning with Applications* 5 (2021), 100036.
- [5] Muhammad Attique Khan, Muhammad Sharif, Talha Akram, Seifedine Kadry, and Ching-Hsien Hsu. 2022. A two-stream deep neural network-based intelligent system for complex skin cancer types classification. *International Journal of Intelligent Systems* 37, 12 (2022), 10621–10649. <https://doi.org/10.1002/int.22691> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/int.22691>
- [6] Marin Benčević, Irena Galić, Marija Habijan, and Danilo Babin. 2021. Training on polar image transformations improves biomedical image segmentation. *IEEE Access* 9 (2021), 133365–133375.
- [7] M.Emre Celebi, Hitoshi Iyatomi, Gerald Schaefer, and William V. Stoecker. 2009. Lesion border detection in dermoscopy images. *Computerized Medical Imaging and Graphics* 33, 2 (2009), 148–153. <https://doi.org/10.1016/j.compmedimag.2008.11.002>
- [8] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*. 248–255. <https://doi.org/10.1109/CVPR.2009.5206848>
- [9] Ta Yang Goh, Shafriza Nisha Basah, Haniza Yazid, Muhammad Juhairi Aziz Safar, and Fathinul Syahir Ahmad Saad. 2018. Performance analysis of image thresholding: Otsu technique. *Measurement* 114 (2018), 298–307. <https://doi.org/10.1016/j.measurement.2017.09.052>
- [10] Simon Graham and Nasir M Rajpoot. 2018. SAMS-NET: Stain-aware multi-scale network for instance-based nuclei segmentation in histology images. In *2018 IEEE 15th international symposium on biomedical imaging (ISBI 2018)*. IEEE, 590–594.
- [11] Md Kamrul Hasan, Md Toufick E Elahi, Md Ashrafur Alam, Md Tasnim Jawad, and Robert Marti. 2022. DermoExpert: Skin lesion classification using a hybrid convolutional neural network through segmentation, transfer learning, and augmentation. *Informatics in Medicine Unlocked* 28 (2022), 100819.
- [12] Xinzhi He, Ee-Leng Tan, Hanwen Bi, Xuzhe Zhang, Shijie Zhao, and Baiying Lei. 2022. Fully transformer network for skin lesion analysis. *Medical Image Analysis* 77 (2022), 102357.
- [13] Long Hoang, Suk-Hwan Lee, Eung-Joo Lee, and Ki-Ryong Kwon. 2022. Multiclass skin lesion classification using a novel lightweight deep learning framework for smart healthcare. *Applied Sciences* 12, 5 (2022), 2677.
- [14] Khalid M Hosny, Mohamed A Kassem, and Mohamed M Fouad. 2020. Classification of skin lesions into seven classes using transfer learning with AlexNet. *Journal of digital imaging* 33 (2020), 1325–1334.
- [15] Aman Kamboj et al. 2018. A color-based approach for melanoma skin cancer detection. In *2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC)*. IEEE, 508–513.
- [16] Muhammad Attique Khan, Muhammad Sharif, Talha Akram, Robertas Damaševičius, and Rytis Maskeliūnas. 2021. Skin lesion segmentation and multiclass classification using deep learning features and improved moth flame optimization. *Diagnostics* 11, 5 (2021), 811.
- [17] Amirreza Mahbod, Gerald Schaefer, Chunliang Wang, Georg Dorfner, Rupert Ecker, and Isabella Ellinger. 2020. Transfer learning using a multi-scale and multi-network ensemble for skin lesion classification. *Computer methods and programs in biomedicine* 193 (2020), 105475.
- [18] Amirreza Mahbod, Philipp Tschandl, Georg Langs, Rupert Ecker, and Isabella Ellinger. 2020. The effects of skin lesion segmentation on the performance of dermatoscopic image classification. *Computer Methods and Programs in Biomedicine* 197 (2020), 105725.
- [19] Scott W. Menzies and Iris Zalaudek. 2006. Why Perform Dermoscopy?: The Evidence for Its Role in the Routine Management of Pigmented Skin Lesions. *Archives of Dermatology* 142, 9 (09 2006), 1211–1212. <https://doi.org/10.1001/archderm.142.9.1211>
- [20] Vigneswaran Narayanamurthy, P. Padmapriya, A. Noorasafin, B. Pooja, K. Hema, Al'aina Yuhainis Firus Khan, K. Nithyakalyani, and Fahmi Samsuri. 2018. Skin cancer detection using non-invasive techniques. *RSC Adv.* 8 (2018), 28095–28130. Issue 49. <https://doi.org/10.1039/C8RA04164D>
- [21] Kishwer S Nehal and Christopher K Bichakjian. 2018. Update on keratinocyte carcinomas. *New England Journal of Medicine* 379, 4 (2018), 363–374.
- [22] Roberta B Oliveira, Joao P Papa, Aledir S Pereira, and Joao Manuel RS Tavares. 2018. Computational methods for pigmented skin lesion classification in images: review and future trends. *Neural Computing and Applications* 29 (2018), 613–636.
- [23] Howard W. Rogers, Martin A. Weinstock, Steven R. Feldman, and Brett M. Coldiron. 2015. Incidence Estimate of Nonmelanoma Skin Cancer (Keratinocyte Carcinomas) in the US Population, 2012. *JAMA Dermatology* 151, 10 (10 2015), 1081–1086. <https://doi.org/10.1001/jamadermatol.2015.1187>
- [24] Caglar Senaras and Metin Nafi Gurcan. 2018. Deep learning for medical image analysis. *Journal of pathology informatics* 9 (2018).
- [25] Connor Shorten and Taghi M Khoshgohar. 2019. A survey on image data augmentation for deep learning. *Journal of big data* 6, 1 (2019), 1–48.
- [26] Rebecca L. Siegel, Kimberly D. Miller, and Ahmedin Jemal. 2020. Cancer statistics, 2020. *CA: A Cancer Journal for Clinicians* 70, 1 (2020), 7–30. <https://doi.org/10.3322/caac.21590> arXiv:<https://acsjournals.onlinelibrary.wiley.com/doi/pdf/10.3322/caac.21590>
- [27] Rajeev Kumar Singh, Rohan Gorantla, Sai Giridhar Rao Allada, and Pratap Narra. 2022. SkiNet: A deep learning framework for skin lesion diagnosis with uncertainty estimation and explainability. *Plos one* 17, 10 (2022), e0276836.
- [28] Parvathaneni Naga Srinivasu, Jalluri Gnana SivaSai, Muhammad Fazal Ijaz, Akash Kumar Bhoi, Wonjoon Kim, and James Jin Kang. 2021. Classification of skin disease using deep learning neural networks with MobileNet V2 and LSTM. *Sensors* 21, 8 (2021), 2852.
- [29] Magnus Thörn, Fredrik Ponté, Reinhold Bergström, Pär Sparén, and Hans-Olov Adami. 1994. Clinical and Histopathologic Predictors of Survival in Patients With Malignant Melanoma: a Population-Based Study in Sweden. *JNCI: Journal of the National Cancer Institute* 86, 10 (05 1994), 761–769. <https://doi.org/10.1093/jnci/86.10.761>
- [30] Mai Thanh Nhat Truong and Sanghoon Kim. 2018. Automatic image thresholding using Otsu's method and entropy weighting scheme for surface defect detection. *Soft Computing* 22 (2018), 4197–4203.
- [31] Philipp Tschandl, Noel Codella, Bengü Nisa Akay, Giuseppe Argenziano, Ralph P Braun, Horacio Cabo, David Gutman, Allan Halpern, Brian Helba, Rainer Hofmann-Wellenhof, et al. 2019. Comparison of the accuracy of human readers versus machine-learning algorithms for pigmented skin lesion classification: an open, web-based, international, diagnostic study. *The lancet oncology* 20, 7 (2019), 938–947.
- [32] Quoc Dang Vu, Simon Graham, Tahsin Kurc, Minh Nguyen Nhat To, Muhammad Shaban, Talha Qaiser, Navid Alemi Koohbanani, Syed Ali Khurram, Jayashree Kalpathy-Cramer, Tianhao Zhao, et al. 2019. Methods for segmentation and classification of digital microscopy tissue images. *Frontiers in bioengineering and biotechnology* (2019), 53.
- [33] Klaus Wolff, Klaus Wolff, Richard Johnson, and McGraw Hill Professional. 2009-03-13. *Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology*. 1104–pages.
- [34] P. H. Youl, B. A. Raasch, M. Janda, and J. F. Aitken. 2007. The effect of an educational programme to improve the skills of general practitioners in diagnosing melanocytic/pigmented lesions. *Clinical and Experimental Dermatology* 32, 4 (07 2007), 365–370. <https://doi.org/10.1111/j.1365-2230.2007.02414.x> arXiv:<https://academic.oup.com/ced/article-pdf/32/4/365/46825795/ced0365.pdf>
- [35] Yushan Zheng, Zhiguo Jiang, Fengying Xie, Haopeng Zhang, Yibing Ma, Huaqiang Shi, and Yu Zhao. 2017. Feature extraction from histopathological images based on nucleus-guided convolutional neural network for breast lesion classification. *Pattern Recognition* 71 (2017), 14–25.